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Rheological properties of Salecan as a new source of thickening agent

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A B S T R A C T

Salecan is a novel soluble glucan produced by *Agrobacterium* sp. ZX09, displaying the ability to inhibit pancreatic amylase and reduce postprandial glucose. The research here provides an investigation of the rheological properties of Salecan solution over a wide range of shear rate (0.001–1000 s^{-1}), frequency (0.1–100 rad/s), concentrations (0.3%, 0.5%, 1.0% and 1.5%), temperature (5–95 °C) and pH (1.0–13.0). The power law model well described the rheological behavior of the solutions, in the shear-thinning region, with high determination coefficients, R^2. Salecan solutions showed a non-Newtonian viscosity behavior at all concentrations and temperatures. The solutions exhibited excellent stability below 55 °C. Viscosities were not affected after being frozen (−20 °C). Over a wide pH range from 6.0 to 12.0, the viscosity almost kept invariant. With increasing frequency, the storage and loss moduli G' and G'' of Salecan solution increased and complex viscosities decreased continuously, which showed an elastic behaviour. All data indicated that Salecan has excellent rheological properties and could be utilized in food industry as a new source of thickening agent.

1. Introduction

Many polysaccharides are hydrocolloids. Because of their ability to modify the rheological and functional properties of food systems, polysaccharides are wildly used in the food industry for gel formation, water retention, emulsification and aroma retention. Food products, such as bread, sauces, syrup, ice cream, instant food, beverages and ketchup, might have hydrocolloids in their formulations (Koocheki, Mortazavi, Shahidi, Razavi, & Taherian, 2009). Starch, one of the most important functional food biopolymers, is added to many products including a variety of low-fat products as a functional ingredient to improve their rheological characters (Gouvêa et al., 2009; Hermansson & Svegmark, 1996).

A variety of microorganisms including lactic acid bacteria, algae, fungi and plants have an ability to produce extracellular polysaccharides and excrete them out of cell either as soluble or insoluble polymers (Wang, Ahmed, Feng, Li, & Song, 2008). Polysaccharides of microbial origin have been developed as source of additives for several industries including xanthan from *Xanthomonas campestris* (Viturawong, Achayuthakan, & Suphantharika, 2008), Succinoglycan from *Sinorhizobium* or *Agrobacterium* (Simsek, Mert, Campanella, & Reuhs, 2009) and gellan from *Pseudomona elodea* (Tako, Sakae, & Nakamura, 1989). Polysaccharides from microbial sources are in particular used as the food additive because they also have the advantage of being regarded as totally natural for many consumers (Lai, Tung, & Lin, 2000). In the aspects of a natural bacterial polysaccharide hydrocolloid, the β-glucan has outstanding functional and nutritional properties. Otherwise, it is attracting the increasing attention of pharmaceutical and functional food industry, not only because of its thickening or gelling properties but also its multiple beneficial effects on human and animal health, such as immune-stimulation (Weintraub, 2003), anti-inflammatory (Dore et al., 2007), antimicrobial (Smiderle et al., 2008), antitumoural (Leung, Liu, Koon, & Fung, 2006), antioxidation (Toklu et al., 2006), cholesterol lowering (Kalra & Jood, 2000) and antidiabetic (Jenkins, Jenkins, Zdravkovic, Würsch, & Vuksan, 2002). It is also reported that β-glucan is nutritionally non-functional in the human digestive tract, and hence has no caloric value (Charalampopoulos, Wang, Pandiella, & Webb, 2002; Morgan, 2000). Viscosity of β-glucan was stable over a wide range of pH and decreased with increasing temperature (Dawkins & Nnanna, 1995). People found that 1.0% (w/w) β-glucan solutions have a low flow behaviour index and a high consistency coefficient in the power law model (Autio, Myllymäki, & Mälkki, 1987).

The molecular weight, structural features and concentration of β-glucans are important determinants of their physical properties, such as water solubility and rheological behaviour, and thereby affect the functionality of food systems. Curdlan is an essentially linear (1→3)-β-glucan, which may have a few intra- or inter-chain (1→6)-linkages, produced from bacterial fermentation. Its aqueous

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suspension can form a gel upon heating, but it is not soluble in water at room temperature (McIntosh, Stone, & Stanisich, 2005). Yeast β-glucan is a fungal β-glucan extracted from yeast cell walls. However, it also has low solubility in aqueous media (Lee et al., 2001). The physical properties of these polymers make these glucans not suited for all applications as food additive.

Despite many studies on the rheological properties of other hydrocolloids, there are very a few reports on the water-soluble linear β-glucan from bacterium. *Agrobacterium* sp. ZX09 could synthesize a lot of extracellular polysaccharide named Salecan which is a novel soluble glucan. It has been proven that Salecan is a liner (1→3)-D-glucan comprising β-1-3-linked glucopyranosyls with small numbers of α-1-3-linked (Xiu et al., 2010). Its molecules with relatively high molecular weight produce viscous solutions which may be related to attenuation of postprandial plasma glucose in fasting mice.

The purpose of this work is to first evaluate the rheological behavior of Salecan in solution, as a new source of hydrocolloid gum used in food. Steady flow and viscoelastic measurements were used. The characterization covers a broad range of shear rate, concentrations, temperatures, solution pH and frequency.

2. Materials and methods

2.1. Salecan extraction and solution preparation

The strain *Agrobacterim* sp. ZX09 used in this study was isolated from the soil sample from ocean coast of Shandong, China. It was inoculated and cultured according to previously described methods (Xiu et al., 2010). The culture broth was added to two volumes of 95% ethanol. Productivity of Salecan was expressed in terms of the weight after ethanol precipitation collected by centrifugation at 6000 g for 15 min and dried under reduced press. Solutions were prepared at concentrations of 0.3%, 0.5%, 1.0% and 1.5% (w/w) by dispersing the required amount of Salecan in deionized water (Milli-Q, Millipore, Bedford, USA), under slow stirring at room temperature overnight prior to assessment.

2.2. Rheological measurement of steady flow properties

Rheological measurements were carried out using a rheometer (Physica MCR 301, Anton Paar, Austria) with a plate geometry sensor (0.05 mm gap). For each test, approximately 2–3 ml sample (0.3%, 0.5%, 1.0% or 1.5%) was placed in the rheometer and maintained for 10 min following by 3 min pre-shearing at 100 s^{-1} to obtain uniform solution. The viscosity was recorded by increasing the shear rate from 0.001 to 1000 s^{-1} at 25 °C. The instrument was programmed to set temperature and equilibrate for 10 min followed by two-cycle shear in which the shear rate was increased linearly from 0 to 300 s^{-1} in 3 min (upward flow curve) and immediately decreased to 0 s^{-1} in the next 3 min (downward flow curve) at 5, 25, 45 or 65 °C. The flow behavior index (n) and consistency index (k) values were computed by fitting the power law model (Eq. (1)):

$$\tau = k\cdot\gamma^{n} \tag{1}$$

where τ is the shear stress (Pa), γ is the shear rate (s^{-1}), k is the consistency coefficient (Pa s^{n}) and n is the flow behavior index (dimensionless). Keeping the shear rate at 0.01 s^{-1} and 50 s^{-1}, apparent viscosity was recorded by increasing the temperature from 5 to 95 °C at a heating rate of 5 °C/min. To study the stability after heating and freezing, the hydrocolloid solutions were first measured at 25 °C. They were then heated to 100 °C or cooled to −20 °C and held for 1 h. The temperature was then returned to 25 °C and the viscosity measured again from 0.1 to 1000 s^{-1}. The viscosity and flow properties were measured for 1.0% gum solutions with pH ranging from 1.0 to 13.0 (1.0 increment, adjusted using 0.1 mol/l NaOH or HCl) at shear rate of 50 s^{-1} and constant temperature of 25 °C.

2.3. Determination of dynamic viscoelastic flow properties

Dynamic viscoelasticity of the prepared Salecan solutions at 0.3%, 0.5%, 1.0% or 1.5% (w/w) were determined. About 2–3 ml sample was placed in the rheometer which was equilibrated to 25 °C. Two steps of rheological measurements were performed: (1) deformation sweeps at a constant frequency ($f = 1$ Hz) to determine the maximum deformation attainable by a sample in the linear viscoelastic range and (2) frequency sweeps over a range of 0.1–100 rad/s (1 Hz = 6.28 rad/s) at a constant deformation (0.5% strain) within the linear viscoelastic range. The storage modulus (G') loss modulus (G''), loss factor ($\tan\delta = G''/G'$) and complex viscosity (η^*) as a function of frequency (ω) were obtained. The generalized Maxwell model was used to fit the frequency sweep data by the method described by Baumgaertel and Winter (1989).

2.4. Statistical analysis

All the measurements were made in triplicate for each sample. One-way analyses of variance followed by Tukey's multiple comparison tests were carried out to study the power law model. The data were obtained using the *software* SPSS version 13.0, $P < 0.05$ was considered as a statistically significant difference.

3. Results and discussion

3.1. Effect of concentration and temperature on flow properties

Fig. 1 shows the effect of shear rate (γ) on viscosity (η) of Salecan at different concentrations at 25 °C. All samples developed non-Newtonian pseudoplastic behavior from our test. Newtonian plateaus for these solutions are observed at extremely low shear rate. The viscosity decreased with increasing shear rate and, as usual as many polysaccharide solutions, the viscosity change was smoothened at high shear rates after a sharp reduction. This shear-thinning behavior was more pronounced as the concentration increased. Linear and stiff molecules have a large hydrodynamic size, which contributes to high viscosity and pseudoplasticity. Most pseudoplastic fluids exhibit Newtonian properties at low shear rate. The shear rate that makes the transition from the Newtonian plateau to shear-thinning region is defined as critical shear rate (γ_c)

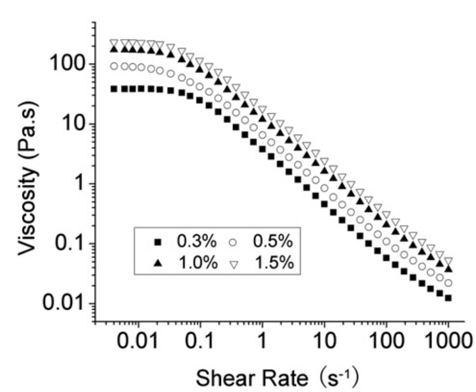

Fig. 1. Viscosity of Salecan solutions with different concentrations at the temperature of 25 °C.

(Fijan et al., 2009). It can be seen clearly on the flow curves that, at each Salecan concentration, γ_c is below 0.1 s^{-1} and lower than many other polymers (Fijan et al., 2009; Launay, Cuvelier, & Martinez-Reyes, 1997). Xanthan gum also shows very low critical shear rate, which was attributed to their propensity to be orientated under shear on account of their extended and semi-rigid conformation (Launay et al., 1997). In the test solutions, the macromolecule of Salecan may show considerable stiffness which makes the chains more easily oriented under shear. Shear thinning is the result of an orientation effect of macromolecules (Vardhanabhuti & Ikeda, 2006). Increasing the shear rate, the long and randomly positioned chains of polymer molecules become aligned in the direction of flow and result in less interaction between adjacent polymer chains (Koocheki et al., 2009). The concentration of polysaccharide in solution is known to affect directly the viscosity and the degree of pseudoplasticity (Sutherland, 1994). The viscosity of many random-coil polysaccharides was shear rate-dependent at high concentrations, while Newtonian flow behaviour was observed at low concentrations (Brummer, Cui, & Wang, 2003; Xu, Willför, Holmlund, & Holmbom, 2009).

Power law model (Eq. (1)) is often used to describe the flow behavior of liquid. In the region of shear thinning, the shear stress (τ) versus shear rate (γ) data for Salecan at different concentrations and temperature fitted well to the power law model with high determination coefficients ($R^2 \geq 0.98$). The consistency coefficient (k) and flow behavior index (n) values as influences of Salecan concentrations and temperatures are shown in Table 1. Increasing Salecan concentration decreased the flow behavior index values (n) and increased the consistency coefficient (k), which are in agreement with the studies of other polymers (Farhoosh & Riazi, 2007; Koocheki et al., 2009; Vardhanabhuti & Ikeda, 2006). Furthermore, no obvious differences were found between the flow behaviors indices (n) and the consistency coefficient (k) of the upward and downward curves. The n value is sometimes called the power law index where $n = 1$ corresponds to Newtonian fluid and the lower n value ($n < 1$) reflects a higher degree of pseudoplastic properties of fluid (Pongsawatmanit, Temsiripong, Ikeda, & Nishinari, 2006). The consistency coefficient k and viscosity η are proportionally related parameters according to the power law model (Dautant, Simancas, Sandoval, & Müller, 2007), and the increase of k values relates to the increase of water binding capacity. The higher viscosity of Salecan solution corresponds to a higher degree of pseudoplastic properties and a lower flow behavior index n. Thus, increasing Salecan solution concentration increased the consistency coefficient k and decreased the flow behaviour index n.

Non-Newtonian behavior became important when the flow behavior index is less than 0.6 (Chhinnan, McWaters, & Rao, 1985; Muller, Pain, & Villon, 1994). It has been reported that gum solutions with high value of flow behavior index (n) tends to feel slimy in the mouth (Szczesniak & Farkas, 1962). Thus, for a provision of a high viscosity and good mouth feel, hydrocolloid characterized by a low n value would be required (Marcotte, Taherian, & Ramaswamy, 2001). Comparatively, the flow behavior index (n) of Salecan was similar with xanthan gum (Vardhanabhuti & Ikeda, 2006), but was lower than many other food additives such as guar gum, monoi gum, *Alyssum homolocarpum* seed gum, CMC (Carboxy Methyl Cellulose), salep, carrageenan, pectin and starch at similar concentrations and temperatures, indicating that this gum is more pseudoplastic (Farhoosh & Riazi, 2007; Koocheki et al., 2009; Vardhanabhuti & Ikeda, 2006).

3.2. Effect of temperature

Effects of temperature on hydrocolloid viscosity vary depending on species of hydrocolloid. The temperature could drive molecular arrangement transition of hydrocolloids. The change of solution viscosity with temperature for xanthan solution is accompanied by first an increase and then a decrease as the temperature ranging from 40 to 60 °C (Rochefort & Middleman, 1987), while increasing temperature from 20 to 80 °C results in a drop of the viscosity of galactomannan by 50% (Wielinga, 2000). Temperature dependency of viscosity of Salecan was also examined.

Table 1
The power law parameters for Salecan solutions at different concentrations and temperatures.

Temperature (°C)	Upward curve			Downward curve		
Concentration	k	n	R^2	k	n	R^2
0.3%						
5	2.50 ± 0.51	0.23 ± 0.03	0.98	2.37 ± 0.48	0.24 ± 0.03	0.99
25	2.79 ± 0.70	0.19 ± 0.03	0.99	2.48 ± 0.64	0.21 ± 0.03	0.99
45	2.81 ± 0.90	0.18 ± 0.04	0.99	2.55 ± 1.34	0.20 ± 0.07	0.99
65	0.14 ± 0.09	0.63 ± 0.07[a]	0.99	0.05 ± 0.04	0.82 ± 0.09[a]	0.99
0.5%						
5	4.77 ± 0.26	0.21 ± 0.01	0.98	4.48 ± 0.23	0.22 ± 0.01	0.99
25	5.28 ± 0.30	0.18 ± 0.01	0.99	4.69 ± 0.34	0.20 ± 0.01	0.99
45	5.41 ± 0.61	0.16 ± 0.02	0.99	4.97 ± 0.44	0.18 ± 0.02	0.99
65	0.24 ± 0.13[a]	0.61 ± 0.07[a]	0.99	0.13 ± 0.08[a]	0.71 ± 0.07[a]	0.99
1.0%						
5	12.32 ± 2.67	0.22 ± 0.02	0.99	11.50 ± 2.44	0.23 ± 0.02	0.99
25	13.53 ± 2.91	0.19 ± 0.02	0.99	12.85 ± 2.88	0.20 ± 0.02	0.98
45	14.62 ± 3.14	0.17 ± 0.02	0.99	13.89 ± 3.27	0.17 ± 0.02	0.99
65	1.96 ± 0.85[b]	0.43 ± 0.08[a]	0.99	0.91 ± 0.52[b]	0.58 ± 0.07[a]	0.98
1.5%						
5	14.47 ± 2.64	0.23 ± 0.02	0.99	12.98 ± 2.15	0.25 ± 0.02	0.99
25	15.62 ± 2.90	0.19 ± 0.02	0.99	14.41 ± 2.39	0.21 ± 0.02	0.99
45	16.65 ± 2.63	0.17 ± 0.01	0.99	15.49 ± 2.55	0.18 ± 0.01	0.99
65	4.30 ± 3.06	0.44 ± 0.10[b]	0.98	2.46 ± 1.55[a]	0.51 ± 0.08[a]	0.98

All the values are means ± SE of 3 replicates.
[a] Values indicate significant differences ($p < 0.05$) with the other three data (5, 25 and 45 °C) in the same column at the same concentration.
[b] Values indicate significant differences ($p < 0.05$) with the data at 25 and 45 °C in the same column at the same concentration.

Wanchoo, Sharma, and Bansal (1996) reported that the consistency coefficient k is a strong function of the temperature and the concentration of the solution. Consistency coefficient (k) and flow behavior index (n) of Salecan solutions as a function of temperature were obtained from the power law model, as shown in Table 1. The effect of temperature on power law parameters was negligible below 65 °C at all concentrations ($p > 0.05$). There is a distinct decline of the consistency coefficient k and an increase of behavior index n, at the temperature of 65 °C, indicating the decrease of the viscosity. The data in Fig. 2A, which have shown the changes of viscosity for 1.0% Salecan solution as an influence of temperature (5 °C, 25 °C, 45 °C and 65 °C), were in accord with those in Table 1. Similar results were obtained for the other concentrations.

To study the stability of Salecan in extremes of temperature, the viscosities of the gum solution at different concentrations were tested from 5 to 95 °C. As shown in Fig. 2B, at the shear rate of 0.01 s^{-1}, there are three distinct flow regions. The first region, at low temperature up to 50 °C (0.3% and 0.5%) or 55 °C (1.0% and 1.5%), is where the viscosity remains constant or increases a little in spite of the increase of temperature. The flow curves in the second region exhibit similar patterns at all concentrations: a sharp decrease of viscosity is observed with the increase of temperature from 55 to 65 °C. In the third region, the viscosity decreases slowly and tends to keep constant, and for solutions at low concentrations (0.3% and 0.5%) there even no viscosity existed. Similar results were also found at the shear rate of 50 s^{-1} except that all samples kept stable low viscosity at high temperature (Fig. 5C). This effect is reversible and it is due to the interactions of the molecules in solution which vary with the temperature. Hassan and Hobani (1998) reported that the viscosity of a solution is a function of the intermolecular forces and water solute interactions that restrict the molecular motion. Therefore, as temperature increases, the thermal energy of the molecules varies and the intermolecular distances change. The critical phenomena theory could explain the viscosity changes of Salecan solution with the increase of temperature. Critical phenomena are those which occur exactly in the phase transition or asymptotically close to it (Stauffer, Coniglio, & Adam, 1982). In this theory, the relation of viscosity η and conversion factor p is $\eta \propto (p_c - p)^{-k}$, where p_c is the critical value and k is the sol viscosity exponent having the values of 0.7 and/or 1.3 depending on the model chosen. Generally, there is a sharp phase transition at some intermediate critical point $p = p_c$ (Stauffer et al., 1982). The value of p can be changed with the temperature increasing, and a sudden decrease of viscosity was observed.

It has been found that xanthan gum could maintain viscosity from 5 to 40 °C (Whitcomb & Macosko, 1978). However, the constant flow behavior of Salecan has extended to 55 °C. There is also a great difference in the temperature dependence of Salecan compared to other gums used in food industry, such as durian seed gum, guar gum and locust bean gum, the viscosity of which decrease with heating (Amin, Ahmad, Yin, Yahya, & Ibrahim, 2007).

The effects of heating and freezing on the viscosity of Salecan are shown in Fig. 3. At the concentration of 0.3% and 0.5%, the viscosity of Salecan decreased after heating at low shear rate (Fig. 3A and B). However, it was reversible after heating or freezing at higher concentrations (Fig. 3C and D). Several hydrocolloids at certain concentration, such as xanthan gum, guar gum and monoi leaf hydrocolloid, have shown the excellent stability during freeze-thaw cycling, the property known as freeze-thaw stability (Vardhanabhuti & Ikeda, 2006). Our result suggests that Salecan at concentrations higher than 0.5% may also be utilized in products that require stability after heating or freezing. Future work needs to be done to look at the stability of Salecan after extended or repeated exposure times to heating and freezing.

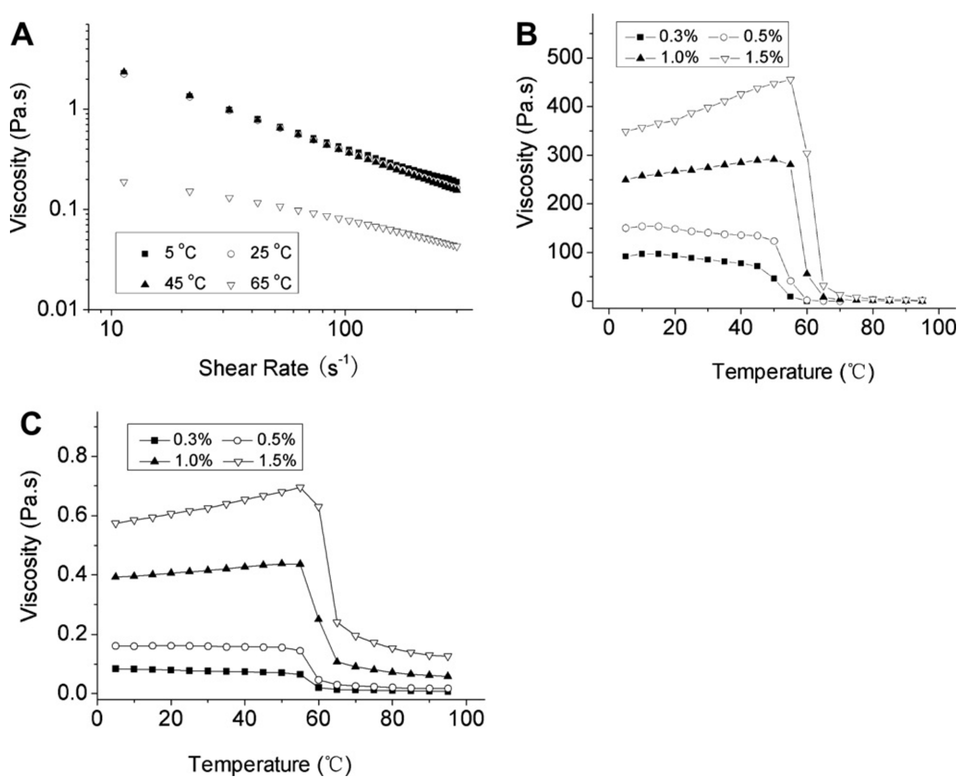

Fig. 2. Effect of temperature on viscosity of Salecan solutions. (A) viscosity of 1.0% Salecan solutions at 5, 25, 45 and 65 °C, the shear rate varied from 0 to 300 s^{-1}; (B) viscosity of Salecan solutions at shear rate 0.01 s^{-1} and temperatures varied from 5 to 95 °C; (B) viscosity of Salecan solutions at shear rate 50 s^{-1} and temperatures varied from 5 to 95 °C.

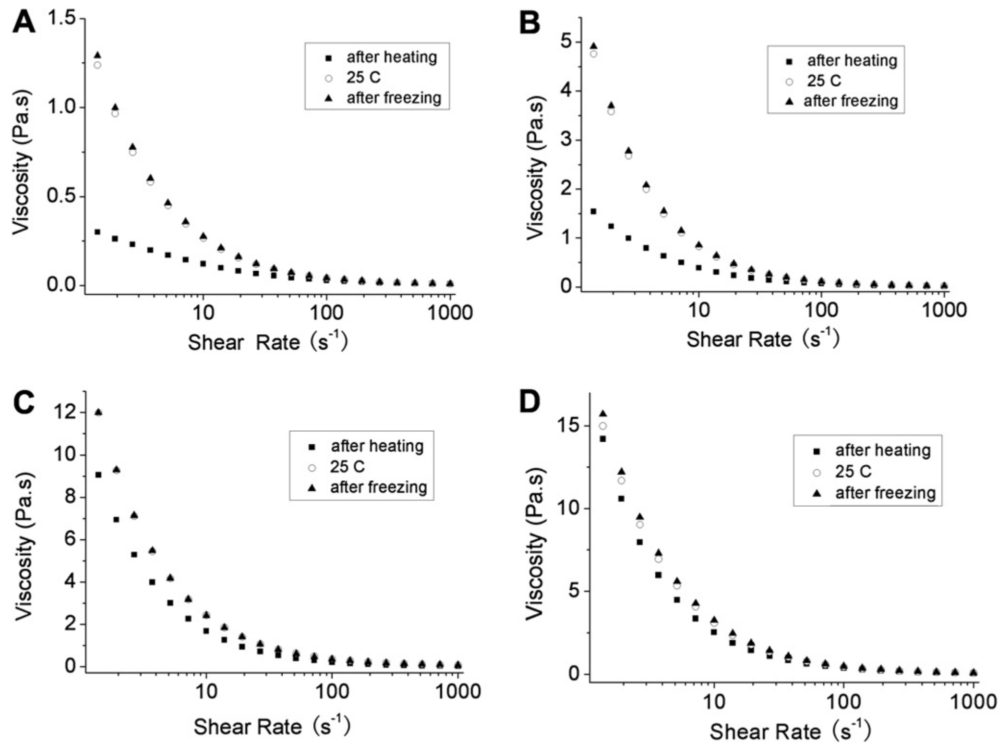

Fig. 3. Effect of freezing and heating on viscosity of Salecan solutions. (A) 0.3% Salecan; (B) 0.5% Salecan; (C) 1.0% Salecan; (D) 1.5% Salecan.

3.3. Effect of pH

The pH value of Salecan solution at room temperature was 6.5–7.5. Changes in pH may influence the viscosity of Salecan. The effect of pH on the steady-shear viscosity (50 s^{-1}) of 1.0% Salecan solution at 1.0–13.0 is shown in Fig. 4. The Salecan solutions were found to be fairly stable over a wide range of pH (6.0–12.0) and increase both at lower and higher pH. Medina-Torres, La Fuente, Torrestiana-Sanchez, and Katthainc (2000) have pointed that the pH effect on the change of apparent viscosity of hydrocolloid was related to conformational changes in the molecule of the polymer. Some hydrocolloids such as xanthan gum and succinoglycan remain stable at low pH (Simsek et al., 2009; Sworn, 2000), while CMC and gum karaya lose their viscosity at the similar pH range (Glicksman, 1982). In addition, galactomannan (William & Phillips, 2000) and guar gum (Wielinga, 2000) can degrade and lose viscosity at high or low pH. It was reported the increase in the intrinsic viscosity of mucilage with the pH (Trachtenberg & Mayer, 1982). Chen and Chen (2001) as well reported that increase in pH, increased the apparent viscosity of the green laver mucilage (*Monostroma nitidium*). It is unusual that the viscosity of the Salecan solutions increases gradually at more acidic or alkaline conditions. These occurrences might be explained as induction of the electrostatic repulsion by functional group that tends to keep the molecules in an extended form in extreme pH, thus producing a high viscous solution (Launay, Doubiler, & Cavalier, 1986; Onweluzo, Obanu, & Onuoha, 1994).

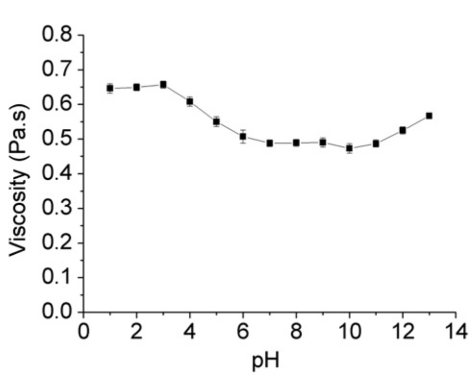

Fig. 4. Viscosity versus pH for 1.0% Salecan solutions at shear rate of 50 s^{-1}. All the values are means ± SE of 3 replicates.

3.4. Dynamic viscoelastic properties

The structure formation of particles or macromolecules in a disperse system can be described by investigating the viscoelastic properties in terms of dynamic measurements in which the elasticity can be described by the storage shear modulus G' and the viscous property can be described by the loss shear modulus G''. The dependence of G' and G'' on frequency can be used to characterize or classify dispersion (Xu, Xu, Zhang, & Zhang, 2008). The slope and the crossover of G' and G'' allow differentiation between a sol/solution, gel-like structure or a dispersion particle system like suspensions, emulsions and foams, and a crossover (G', G'') is for example an indicator of a gel-like structure (Dongowski et al., 2005). Keeping at 0.5% strain, which was determined in the liner regime of viscoelasticity in our test, typical oscillatory flow curves as a function of Salecan concentration in deionized water at 25 °C are shown in Fig. 5. As can be observed, both the dynamic storage modulus G', and the viscous modulus G'' show a dependency on the frequency from 0.1 to 100 rad/s (Fig. 5A). Increasing frequency increased both storage (G') and loss moduli (G''). This behavior is clearly dependent on Salecan concentration. At all concentrations, G' is always superior to G'', and not any crossover between these two moduli was observed throughout the measured frequency range

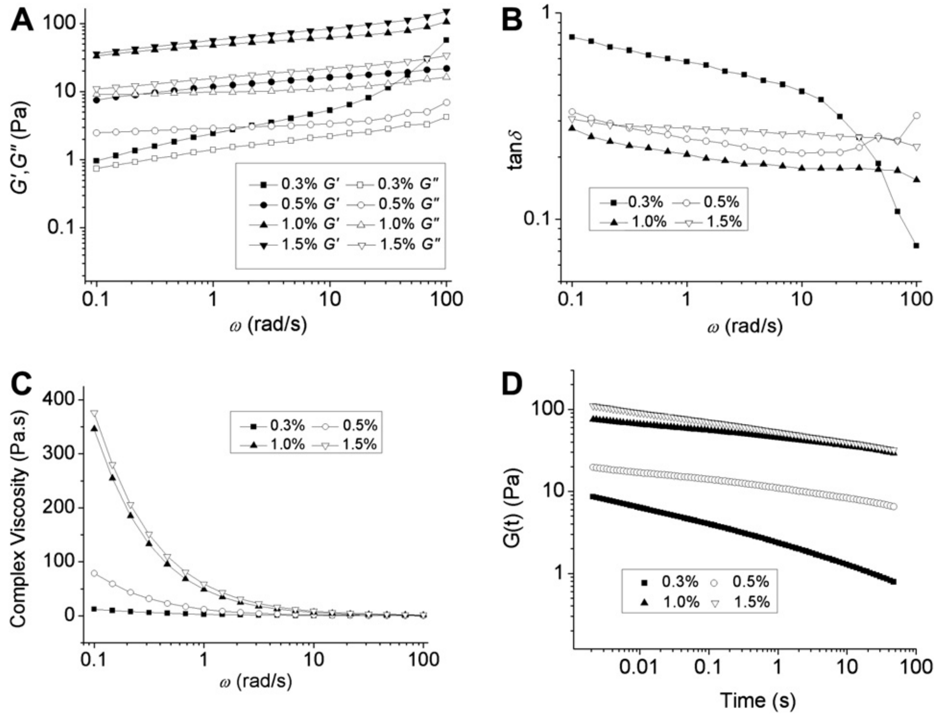

Fig. 5. Dynamic mechanical spectra and relaxation spectrum of Salecan solutions. Measurements were made at 0.5% strain and 25 °C. (A) Dynamic moduli; (B) Loss factor (tanδ); (C) Complex viscosity; (D) Relaxation spectrum.

(Fig. 5A). This means that the elastic behavior dominates over the viscous component throughout the entire frequency range examined. While in oscillation tests of β-glucan preparations from oat products, G' was higher than G'' at low frequencies and G'' was higher at high frequencies where both moduli increased with increasing frequency (Autio, 1988).

On the other hand, increasing Salecan concentration from 0.3% to 0.5%, G' becomes greater than G'' indicating a clear tendency to form macromolecular networks with important elastic properties. It accords with the previous finding that the gelation rate rises with decreasing molar mass and increasing β-glucan concentration (Böhm & Kulicke, 1999).

At the concentration higher than 0.5%, the loss factor tanδ, which is defined as the quotient of G'' and G', almost keep at a constant range (Fig. 5B). The dominant elastic behaviour of solutions was derived from the fact that the storage modulus was higher than the loss modulus. The higher the loss factor, the higher is the proportion of dissipated energy due to viscous flow under external stress. The structural stability of the solutions can also be determined. The fact that the tanδ (G''/G') values for all the samples tested were smaller than unity indicates predominantly elastic behaviour (Fig. 5B). The tanδ decreased, indicating that the elastic behavior increased, with the rise of frequency. In highly viscous solutions, the fluid properties are important as food additive. The overall response to the applied deformation is monitored by the complex viscosity η^* (Madden, Dea, & Steer, 1986), which is the best parameter for comparing the fluidity of the materials. In Fig. 5C, the complex viscosity of the samples, which generally reflect the trends observed in the dynamic moduli, decreased with increasing the frequency. The complex viscosity η^* was also affected by the concentration. At the high concentrations (1.0% and 1.5%), there is a strongly frequency dependence, indicative of a structured material. At the lower concentrations (0.3% and 0.5%), the solutions are much less frequency dependent, and a low frequency viscosity plateau appears (0.3%). This is typical dilute solution behavior which indicates a loss of structure at these conditions.

Calculations of viscoelastic fluids could also be described by relaxation modulus and relaxation time with the Maxwell model. It is a classical problem of rheology, which could be avoided by directly measuring the relaxation modulus $G(t)$ in the time domain, to convert the dynamic data from the frequency domain to the time domain. Generalized Maxwell model was used to process and analysis the dynamic mechanical data. From measured data G' and G'', the relaxation spectrum was calculated (Fig. 5D). Before the gel point, the polymer can relax completely, at the gel point the polymer relaxes in a power law, and beyond the gel point the polymer can only relax to an equilibrium value (Baumgaertel & Winter, 1989). The data shown here indicates that these are systems with predominantly elastic characteristics.

4. Conclusion

This study concluded that aqueous solutions of Salecan exhibited non-Newtonian flow behaviour from 0.3% to 1.5%. It was observed that only a small quantity of Salecan could produce a marked shear thinning. For this reason Salecan will be more appropriate to use as food additive compared with many other polymers. The power law model well described the rheological behavior of the solutions. Increasing solution concentrations increased the viscosity and pseudoplasticity. Temperature had relatively little effect on the viscosity of Salecan solution below 55 °C. Flow behavior remained stable after freezing. The pH almost has no influence on apparent viscosity of Salecan solutions between 6.0 and 12.0. Dynamic measurements also suggested that elastic behavior dominates over the viscous component throughout the entire frequency range examined. It can be seen that Salecan may be utilized in foods to give viscosity, body, and mouth feel as a new source of thickening agent.

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